



Tara Chand

Roll No.: 230104108

B.Tech - Civil Engineering

Indian Institute of Technology, Guwahati

+91-70562 02923

t.chand@iitg.ac.in

tarachandgarg79@gmail.com

tarachand.tech | GitHub | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech, Civil Engineering	Indian Institute of Technology, Guwahati	8.55 (Current)	2023-Present
Senior Secondary	Central Board of Secondary Education	94.4%	2022
Secondary	Board of School Education, Haryana	92.8%	2020

EXPERIENCE

- Outsyde — Adventure Booking Platform** May 2026 – Jul 2026
Full-Stack Developer Intern Remote
 - Architected a **51-table PostgreSQL** schema using **Prisma** supporting bookings, payments, referrals, **audit logging**, and compliance workflows, optimizing relational design for maintainable, highly scalable, and **transaction-safe** backend operations.
 - Engineered a **9-module admin console** with fine-grained **RBAC** and centralized management across **64+ REST APIs**, automating **approval workflows** and substantially cutting manual administrative effort through workflow automation and monitoring.
 - Built the **end-to-end booking platform** with **ACID-compliant** payments, provider and consumer portals, and an idempotent **referral & loyalty engine** using **retry-safe** transaction handling for platform-wide consistency and data integrity.

PROJECTS

- GameHub — Multi-Game Platform** | *Group Project* Mar 2026 – Apr 2026 | Website
 - Architected a **plugin-based multi-game platform** using **Node.js**, **Express**, **MongoDB**, and **Socket.IO**, supporting **5+ games** through one shared engine interface handling scoring, lifecycle, game execution, and runtime configuration management.
 - Built **JWT + bcrypt** authentication with **OTP-based email verification** and email flows across signup, verification, contact.
 - Engineered **server-authoritative** real-time multiplayer via **Socket.IO**, managing room, seat, and turn state across players.
- Smart Traffic Command Center** | *Group Project* May 2026 – Jun 2026 | Website
 - Engineered **64 predictive features** from **8,173** incident records and trained chained **gradient-boosting models**, achieving **ROC-AUC 0.989** and R^2 **0.999** using a **leakage-safe** train/test pipeline with explainable predictions guiding pre-event deployment.
 - Developed the **inference runtime and dashboard** serving predictions and recommendations, deployed on **Streamlit Cloud**.
 - Derived a data-driven severity score via **PCA-weighted feature blending** across 7 disruption indicators for resource allocation.
- TextLite — Terminal Text Editor** | *Personal Project* Jul 2026 – Jul 2026 | Github
 - Built a terminal-based text editor in **C++** using a **custom singly linked list** for in-memory line management, requiring **zero external dependencies** beyond the standard library with modular architecture, memory safety, and unit-tested core logic.
 - Implemented **O(1) insert/remove** after positioning via **pointer-based splicing**, eliminating costly O(n) array-based shifts.
 - Structured a modular, testable architecture with **dual build systems** (Make and CMake) and automated **CI** integration pipelines.
- MathSage — RAG CLI for Math Q&A** | *Personal Project* Jul 2026 – Jul 2026 | Github
 - Developed a **Retrieval-Augmented Generation** pipeline in **Python** using **LangChain**, **Chroma**, and **Gemini**, preserving **multi-turn conversational context** while grounding responses in live Stack Exchange data using vector similarity search.
 - Indexed **30+ documents** per query with **sentence-transformer** embeddings, retrieving the **top-3** candidates for generation.
 - Designed an **LCEL retrieval chain** (retriever, prompt, LLM, parser) for swappable stages and prompt-constrained generation.

TECHNICAL SKILLS

- Languages:** C, C++, Python, JavaScript, TypeScript
- Web & Backend:** React, Next.js, Redux, Tailwind CSS, Node.js, Express, REST APIs, WebSockets/Socket.IO, JWT/OAuth
- Databases & Services:** MySQL, PostgreSQL, MongoDB, Prisma ORM, Firebase
- Data Science & ML:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, LangChain
- Core Fundamentals:** DSA, Object Oriented Programming, DBMS, OS, Probability, Statistics
- Tools & Other:** Git, GitHub, Postman, Docker*, MATLAB*, L^AT_EX* * Elementary proficiency

ACHIEVEMENTS

- Top 75 nationally**, among **30K+** registrations, selected for the final code-a-thon at Cisco campus, Bangalore 2026
- Semifinalist**, Flipkart GRiD 8.0, India's largest engineering challenge, Software Development Track 2026
- Semifinalist**, Flipkart Gridlock 2.0, qualified with the Smart Traffic Command Center project 2026
- Semifinalist**, among **47K+** registrations, MyCareernet HackVega 2.0, national software hackathon 2026
- Top 4.1%**, among **1.8L+** candidates in JEE Advanced 2023
- Rank 2,945**, among **11.6L+** candidates nationwide in the Joint Entrance Examination (Main) 2023
- Global Rank 143**, among **25K+** participants in CodeChef Starters 236 (**3★**, peak rating **1650**) Link
- Pupil at Codeforces**, peak **1376**, solved **1000+** DSA problems across CF, LeetCode, CodeChef & AtCoder Card

POSITION OF RESPONSIBILITY

- Core Team, Public Relations & Branding**, Alcheringa'26 (Cultural Fest), IIT Guwahati 2026
- Organiser, Technothon**, Techniche'24 (Techno-Management Fest), IIT Guwahati 2024

EXTRA-CURRICULAR ACTIVITIES

- Student Mentor**, mentored students in Physics at **Physics Wallah**, guiding JEE aspirants 2024
- Participant, Meta Hacker Cup**, Meta's annual global competitive programming contest 2024
- Participant**, Mime Battle in **Manthan**, Inter-Hostel Cultural Competition, IIT Guwahati 2024